

Today's lecture

So far we discussed:

- ① How is a distribution represented as a DAG? Factorization
- ② How do you draw a BN starting from a distribution?
- ③ How do you read off CIs from a BN? (a) Local-CI (b) Global-CIs via d-separation

Today:

- ① A better algorithm for BN construction.
- ② Limitation of BNs
- ③ Introduction to MRFs.

An optimal algorithm for learning BNs when a BN exists for a distribution (PC Algorithm)

$$\underline{x_i \perp\!\!\!\perp x_j | Pa(x_i), Pa(x_j)} \mid x_i \neq x_j \in G$$

Given: $P(X_1, \dots, X_n)$ that can answer CI queries over any three subsets of variables.

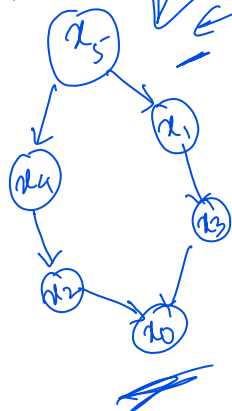
- 1 Initialize G to be a complete graph.
- 2 For each pair of variables x, y , if there exists a subset Z such that $x \perp\!\!\!\perp y | Z$
 - 1 Remove edge between x, y
 - 2 $W(x, y) = Z$ ← Witness
- 3 For each triple x, y, z such that $x \text{---} z$ $y \text{---} z$ and no edge between x, y , and $z \notin W(x, y)$



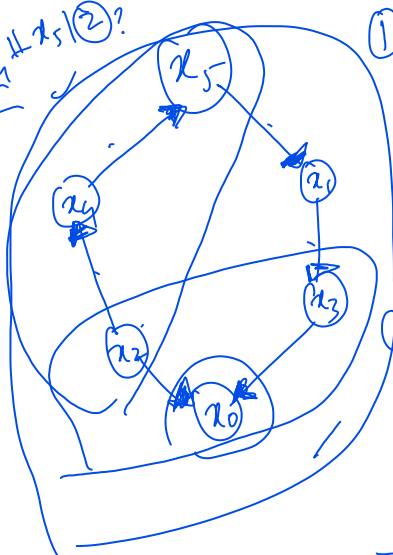
- 1 Orient the edges as $x \rightarrow z \leftarrow y$
- 4 For un-oriented edges, choose edge orientation s.t. no ^{new} immorality is introduced & G is a DAG

Example

Hidden



$x_4 \perp\!\!\!\perp x_5 \mid (2)?$



① $x_5 \perp\!\!\!\perp x_4? No$
 $x_5 \perp\!\!\!\perp x_4 \mid x_1, x_3$
 $x_5 \perp\!\!\!\perp x_4 \mid x_2, x_3$

② $x_5 \perp\!\!\!\perp x_2? No$
 $x_5 \perp\!\!\!\perp x_2 \mid x_1, No$
 $x_5 \perp\!\!\!\perp x_2 \mid x_4$
 $w(x_2, x_5) = x_4$

③ $x_2 \perp\!\!\!\perp x_3 \mid x_5$
 $w(x_2, x_3)$

PC Algorithm is correct only if P can be perfectly represented by a BN

Failure modes of PC

$$x, y, z \in \{0, 1\} \\ P(x) = \frac{1}{2}$$

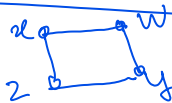
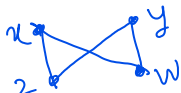
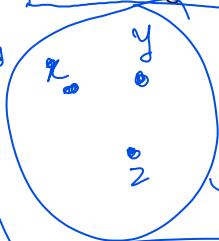
- 1 E.g. $x \perp\!\!\!\perp y, x \perp\!\!\!\perp z, y \perp\!\!\!\perp z, x \not\perp\!\!\!\perp \{z, y\}$
PC algorithm will return (wrongly) a disconnected graph.

$$z = x \oplus y$$

- 2 $x \perp\!\!\!\perp y|z, w \quad z \perp\!\!\!\perp w|x, y$ Not possible to find a DAG without introducing new immoralities.

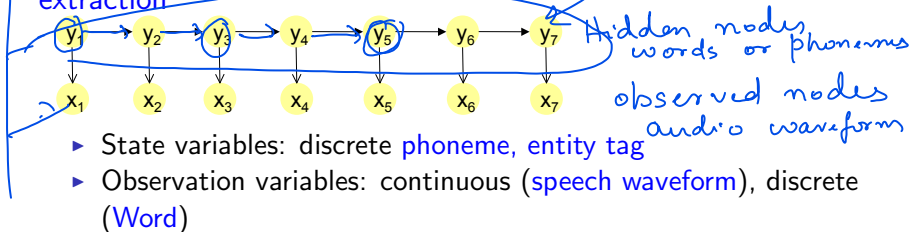
$$\begin{aligned} w(x, y) &= \emptyset \text{ since } x \not\perp\!\!\!\perp y \\ w(y, z) &= \emptyset \quad " \quad y \perp\!\!\!\perp z \\ w(x, z) &= \emptyset \quad " \quad x \perp\!\!\!\perp z \end{aligned}$$

$$x \not\perp\!\!\!\perp \{z, y\}.$$



Popular Bayesian networks

- Hidden Markov Models: speech recognition, information extraction



- Kalman Filters: State variables: continuous
 - Discussed later
- Topic models for text data
 - Principled mechanism to categorize multi-labeled text documents while incorporating priors in a flexible generative framework
 - Application: news tracking
- QMR (Quick Medical Reference) system
- PRMs: Probabilistic relational networks:

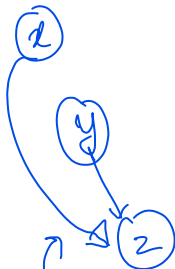
Can All Distributions be Represented as BNs?

$$x \perp\!\!\!\perp y, x \perp\!\!\!\perp z, y \perp\!\!\!\perp z, x \not\perp\!\!\!\perp \{z, y\}$$

Example: $x = y \oplus z$

Involve the ordered insertion algo.

x, y, z



$$\min_S x \perp\!\!\!\perp y \mid S$$

$x \perp\!\!\!\perp z$
 $x \perp\!\!\!\perp y$
 } are lost in G

$$\min_S z \perp\!\!\!\perp \{x, y\} \mid S$$

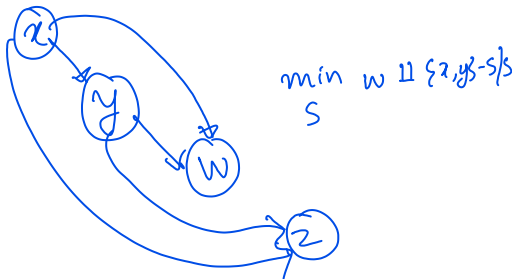
Can All Distributions be Represented as BNs?

Symmetric dependencies: $x \perp\!\!\!\perp y | z, w$ $z \perp\!\!\!\perp w | x, y$

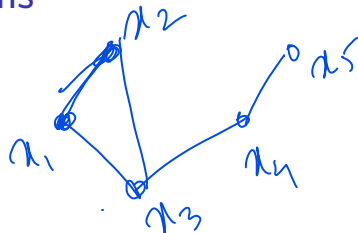
Example: spread of information among symmetric friends

x, y, w, z

$x \perp\!\!\!\perp y | z, w$



Definitions for Undirected Graphs

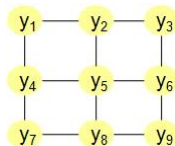


Given undirected graph (V, E)

- 1 Clique: subset C of vertices V where all pairs $(u, v) \in C$ are present in E . $\{x_1, x_2, x_3\}, \{x_1, x_2\}$
- 2 Maximal cliques: subset C such that its superset is not a clique. $\{x_1, x_2, x_3\}, \{x_3, x_4\}, \{x_4, x_5\}$

Undirected graphical models

- Graph G : arbitrary undirected graph
- Useful when variables interact symmetrically, no natural parent-child relationship
- Example: labeling pixels of an image.
- Potentials $\psi_C(\mathbf{y}_C)$ defined on arbitrary cliques C of G .
- $\psi_C(\mathbf{y}_C)$: Any arbitrary non-negative value, cannot be interpreted as probability.
- Probability distribution



$$\Pr(y_1 \dots y_n) = \frac{1}{Z} \prod_{C \in G} \psi_C(\mathbf{y}_C)$$

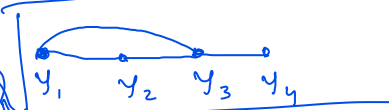
where $Z = \sum_{\mathbf{y}'} \prod_{C \in \text{Cliques}(G)} \psi_C(\mathbf{y}'_C)$ (partition function)

Example 1

Each $y_i \in \{s, i, h, e, t, r, a\}$

$$P(y_1, y_2, y_3, y_4) = \phi_{12}(y_1, y_2) \phi_{23}(y_2, y_3) \phi_{34}(y_3, y_4) \phi_{13}(y_1, y_3)$$

$$\tilde{P}(y_1, y_2, y_3, y_4)$$



$$\phi_{12}(y_1, y_2) =$$

	s	i	h	e	t	r	a
s	0	3	2	1	1	1	2
i							
h							
e							
t							
r							
a							

$$Z = \sum_{y_1, y_2, y_3, y_4 \in \{s, i, h, e, t, r, a\}^4} \tilde{P}(y_1, y_2, y_3, y_4)$$

$$4 \times 7 \times 7$$

$$\tilde{P}(\text{shit}) = \phi_{12}(s, h) \phi_{23}(h, i) \phi_{13}(s, i) \phi_{34}(i, l)$$

$$P(\text{shit}) = \frac{1}{Z} \tilde{P}(\text{shit})$$